α angle of attack (measured between free-stream velocity direction and the model baseline), deg

β angle of sideslip, deg

TESTS

Since the exhaust-heating problem was known to occur at cruise conditions for this helicopter configuration, tests were made for a range of V_{eff} from 0.6 to 1.0. Although tunnel dynamic pressure was as high as 2873 N/m^2 (60 lbf/ft^2), the best visualization of flow patterns was obtained at a dynamic pressure of 211 N/m^2 (4.4 lbf/ft^2). The angle-of-attack range was 0° to -10° and the model sideslip angle range was 10° to -10°.

Five flow deflector configurations were tested, as shown in figure 3. These configurations were (1) a flow deflector with the center cut out so that it would fit down over the entire cowling (fig. 3(a)); two modifications of this configuration are shown in figures 3(b) and 3(c); (2) a flow deflector with outboard sections canted downward (fig. 3(d)); and (3) a spoiler which is a standard installation (figs. 2 and 3(e)) on the gunship helicopter being investigated. The flow deflectors were mounted on two aluminum struts attached to the engine cowling.

APPARATUS

Model

The model was composed of the engine exhaust section attached to a wooden representation of a section of a helicopter pylon and fuselage structure. (See fig. 4.) A photograph of the helicopter exhaust system model installation in the V/STOL tunnel is shown in figure 5. Because the full-scale rotor diameter was greater than the tunnel width, the rotor was not installed. However, it was not considered necessary to have the effects of the rotor for the purposes of this investigation. An analysis of the rotor downwash utilizing the method of reference 8 indicates that at the cruise conditions, the tip vortex will pass high enough above the cowling to cause little influence upon the exhaust exit.

Flow-Visualization Equipment

The primary investigatory equipment was a helium-bubble generator system. This system is useful for airflow visualization when the flow temperature is less than 54° C (130° F) and flow velocity is less than 61 m/sec (200 ft/sec). Bubbles were released near the exhaust and inside the exhaust duct. These bubbles were illuminated by the